

Machine-learning-based prediction of vortex-induced vibration in long-span bridges using limited information

Sunjoong Kim^a, Taeyong Kim^{b,*}

^a Department of Civil Engineering, University of Seoul, Seoul, South Korea

^b Department of Civil and Mineral Engineering, University of Toronto, Toronto, Canada

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ABSTRACT

Long-span bridges are susceptible to wind-induced vibration due to their high flexibility, low-frequency dominance, and light damping capacity. Vortex-induced vibrations (VIVs), which usually occur under in-service conditions, can result in discomfort to users and detrimental effects on the fatigue capacity of structural elements; therefore, accurate VIV assessments are essential in ensuring the vibrational serviceability of bridges. Despite the research efforts of data-driven VIV prediction, the robustness and general applicability of the proposed methods remains challenging, in that each method requires different conditions for the datasets in order to develop machine-learning (ML) models. Furthermore, collecting sufficient VIV datasets (anomaly state) from various operational conditions is impractical, time-consuming, and even impossible in some situations compared with non-VIV datasets (normal state). This imbalance in the dataset could degrade the model performance. To address this issue, this paper focuses on developing a general framework for introducing ML algorithms to predict VIVs with a limited amount of information. To properly replicate the practical cases, two different scenarios are assumed along with the amount of VIV data: (1) no VIV data are available, or (2) only a small number of VIV data can be obtained. A variety of ML-assisted methods are introduced for each scenario to predict VIVs in order to demonstrate the versatility of the proposed framework. The effectiveness and applicability of the proposed framework are demonstrated using actual monitoring data. Different methods are prepared to provide further insight into the ML algorithms used for VIV prediction. The proposed framework in this paper is expected to advance our knowledge and understanding of the application of ML algorithms to bridge systems, which are essential in enhancing resilience against wind hazards.

1. Introduction

Long-span bridges feature the interconnection of two distant locations and support for the mainstreams of logistic movement, which reduces the risk of ship collisions with piers. For decades, the construction of long-span bridges has accelerated worldwide with the development of lightweight materials, computational powers, and construction technologies [1]. Considering the socio-economic benefits, it is important to ensure the normal operational states of long-span bridges. However, due to structural characteristics that include high flexibility, low frequency dominance, and light levels of damping, long-span bridges are generally susceptible to wind-induced vibrations. Indeed, excessive wind-induced vibrations have been reported in many long-span bridges [2-6].

Vortex-induced vibration (VIV) is both single- and multi-modal [7-9] and is a type of wind-induced vibration that is caused by vortex shedding

in the wake that commonly occurs in low wind-speed regions. Thereby, VIV could seriously impact the vibrational serviceability and the long-term fatigue life of bridges. VIVs have been frequently observed on several long-span bridges worldwide, and numerous researchers have reported the detrimental effects on bridges [3,10-14]. Past research efforts have shown three main causes of VIVs [11,15-18]: (1) damping ratios for full-scale bridges that are lower than the design value; (2) variations in the surrounding environmental and operational conditions; and (3) insufficient replication in wind tunnel tests investigating VIVs. Due to the unexpected behaviors of wind forces and traffic conditions as well as to a high degree of uncertainty in analyses, experiments, and construction, the role of long-term monitoring has taken on a greater level of importance.

Machine Learning (ML) or Deep Learning (DL) combined with structural health monitoring (SHM) datasets show promise for the data-

* Corresponding author.

E-mail address: tyong.kim@mail.utoronto.ca (T. Kim).

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driven condition assessments of long-span bridges [17,19–23]. In this regard, several studies have proposed ML/DL-based VIV identification frameworks that can be classified into one of three types in terms of the introduced ML techniques. The first type is the introduction of unsupervised ML models. For instance, Li et al. [21] and Arul et al. [19] introduced clustering analysis to partition a dataset into distinct clusters (VIV and non-VIV) in terms of similarities. The second type is to develop a classification model. Li et al. [22] employed a decision tree algorithm to classify the VIV modes for given wind characteristics, whereas Xu et al. [23] developed a two-stage wind field prediction model similar to a decision tree. Lim et al. [17] trained a deep neural network model to classify either VIV or non-VIV datasets using a semi-supervised labeling process. The third type is to develop a regression model. Using the predicted values from the regression model, VIV is determined by comparing the predicted and the measured values. Huang et al. [20] introduced the random-decrement technique to preprocess the acceleration data and detect VIVs by setting the threshold as the coefficient of variation for a preprocessed dataset. Note that some research efforts have combined various ML techniques to improve the performance of the model and framework, but such studies are classified into one of these three types in terms of a technique that plays a critical role in predicting VIV.

Despite these successful implementations, the performance of ML-based prediction models highly depends on the composition of the dataset. For instance, some ML-based prediction models require both VIV and non-VIV datasets to predict VIVs, but some bridge systems only have a normal-state dataset (i.e., non-VIV). Such a condition is widely observed in real practice considering that VIVs rarely occur compared with normal operations. This results in a case where no VIV dataset is available. In other words, collecting sufficient VIV datasets is often impractical, always time-consuming, and may not even be feasible, which could degrade the model performance and hamper applicability and feasibility in a real-world situation.

To address this issue, this study focuses on proposing a general framework to develop ML-based models to predict VIVs using limited information in terms of different conditions of measured datasets. In this regard, we considered two commonly encountered scenarios in actual practice: (1) no VIV dataset, and (2) imbalanced datasets with extremely small VIV events compared with normal operations. For convenience, these two scenarios will hereafter be denoted as S1 and S2. In the present study, S1 is addressed using the prediction error to detect anomalies or to generate a dataset of abnormal operation states before training the model. On the other hand, S2 is resolved by an augmented VIV dataset; then, ML models are trained with both the measured and augmented datasets. Since S1 is a subset of S2, one can use the first scenario framework even though few VIV data-points are available. The applicability and effectiveness of the framework are demonstrated here by applying it to the Jindo Bridge, a twin cable-stayed bridge in South Korea.

This paper mainly consists of five sections. Section 2 describes the ML-based VIV prediction framework according to the two scenarios. Next, Section 3 presents the examined bridge system and measured datasets. The feasibility of the proposed methodologies is demonstrated in Section 4 by application to the field data obtained from the cable-stayed bridge. In addition, a series of parametric studies is carried out to explain the extracted features from each ML model. A comparison of each methodology is presented to determine the characteristics and difficulties of the application. Finally, a summary and conclusions are presented in Section 5.

2. ML-based VIV prediction

The ML algorithms are generally categorized as falling into one of two ways of learning according to the given datasets: either supervised or unsupervised. Supervised learning can further be classified into either regression or classification according to the handling of either

continuous or discrete output values, respectively. Regarding unsupervised learning, autoencoder and clustering are widely used methods in the practice and research fields — autoencoders detect anomalies and clustering analysis categorizes similar data-points into distinct groups.

Two different scenarios are usually encountered in actual VIV assessments, and these are introduced in this study. S1 represents the lack of an available VIV dataset in training ML models. Such a case could be a recently constructed bridge. On the other hand, a VIV dataset is available in S2, but the number of VIV cases is usually significantly smaller than that of a non-VIV dataset. Details of the framework are described in the subsequent sections to show the applicability of the ML algorithms in predicting VIVs.

2.1. No VIV dataset

A schematic diagram of the proposed framework for S1 is presented in Fig. 1. In Stage 1, key features such as 10-minutes of root mean square (RMS) acceleration, the corresponding mean wind speed, average temperature, number of vehicles, and gross vehicle weight (GVW) are extracted from the collected data. Then, VIV occurrences are labeled based on researchers' knowledge and experience. Once the datasets are manually annotated, only the non-VIV dataset (normal state) is used to train ML models. Next, in Stage 2, the following three different strategies are introduced to predict VIVs: probabilistic regression, data augmentation, and unsupervised learning. Due to the non-availability of the VIV dataset, the ML models were focused on predicting the non-normal state in Stage 2, which means the VIV dataset is a subset of the predicted non-normal state. Thus, in Stage 3, engineering judgments and predefined conditions are used to identify the VIV state from the predicted non-normal state. The conditions and thresholds may vary along with bridge systems and operational conditions.

Among the three ML models in Stage 2 of Fig. 1, a probabilistic regression model could be first developed that combines the environmental and operational conditions (EOCs) and output from the RMS vertical acceleration of the bridge deck. Since the model is trained using only the non-VIV dataset, the predicted responses can only properly describe a normal operational state. In other words, if the EOCs for the non-normal state including VIV datasets are employed as input for the trained model, a large discrepancy could be observed between the predicted and measured values. The confidential interval (CI) is used as a threshold to detect any anomalies against the normal state.

Second, a dataset in the abnormal state (VIV) is randomly augmented based on sampling techniques. To this end, a Gaussian mixture is employed to construct a probability density function (PDF) of the non-normal state dataset. Random samples are then generated using the constructed PDF. A data-point with a probability density that is less than a certain threshold value is stored until the number of generated samples is equivalent to that of a non-VIV dataset (i.e., a normal state). By labeling the normal state (non-VIV) and the abnormal state as 0 and 1, a classification model can be developed for a given set of EOCs. If the model is properly trained using the datasets, an abnormal state can be effectively classified apart from a normal state. Note that (1) as stated before VIV is a subset of the abnormal state, which means that not every abnormal state guarantees VIVs, and (2) any random-generation methods can be used instead of a Gaussian mixture.

Third, among various unsupervised learning methods, an autoencoder can be adopted for S1. Autoencoders are widely used for dimensionality reduction or anomaly detection by sequentially going through the encoder and decoder mapping functions [24]. The encoder facilitates the transformation of a d -dimensional input vector to an r -dimensional latent variable where $r < d$, while the decoder transforms the latent vector back into a reconstructed vector of the d dimension. Since an autoencoder model trained by a dataset of normal conditions cannot effectively reconstruct the data from an abnormal condition, reconstruction errors can be used to measure anomalies. In this regard, the autoencoder model is trained only using a non-VIV dataset; the non-

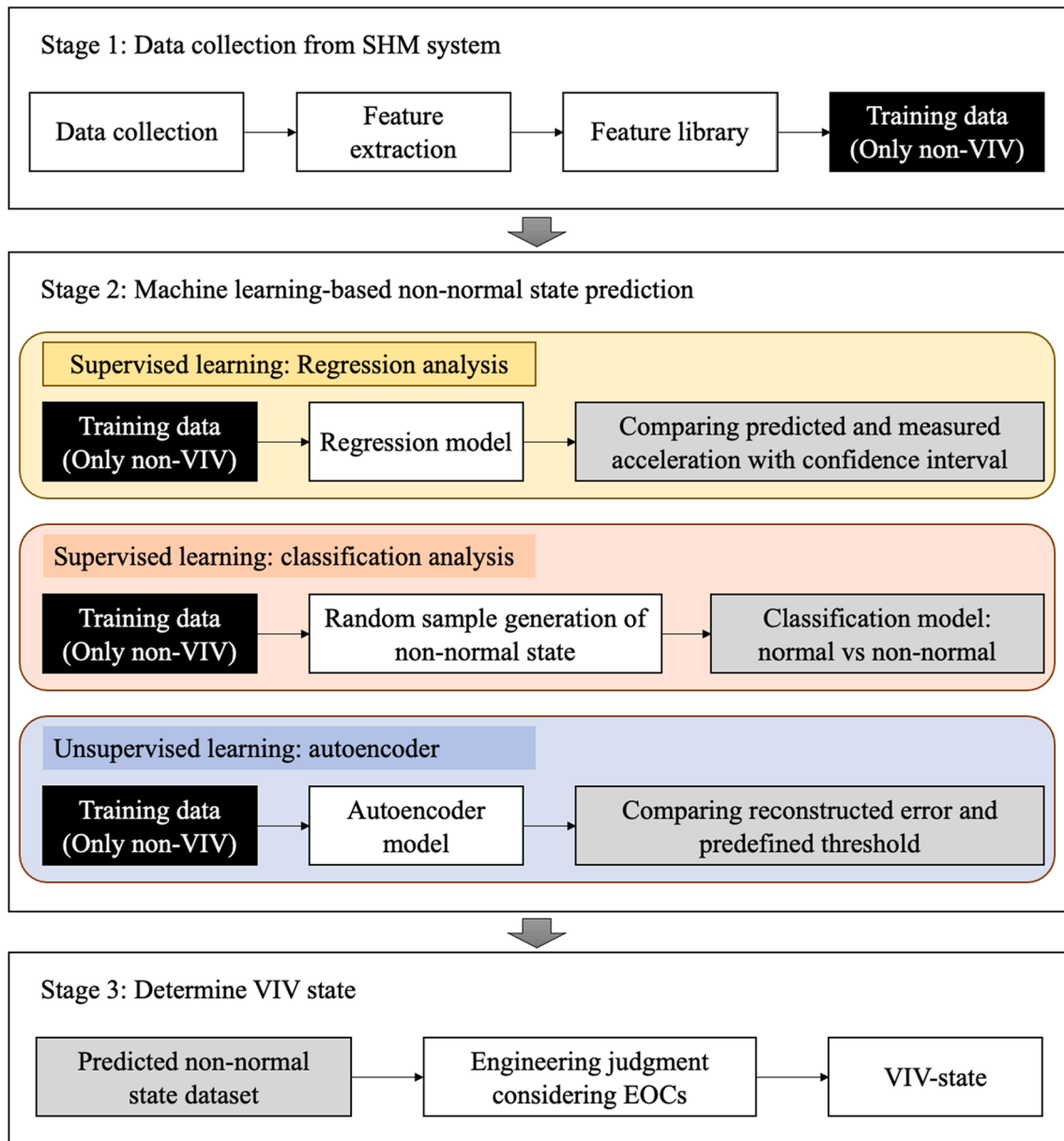


Fig. 1. ML-based VIV prediction when no VIV dataset is available.

normal state including VIV can then be detected by determining whether a reconstruction error exceeds a certain threshold or not.

After predicting the non-normal state from the ML models, the post-processing (Stage 3 in Fig. 1) is carried out to confirm the predicted non-normal states as VIV events. Given the fact that VIV occurs in specific wind-speed regions, it is recommended to decide the predicted anomalies as a VIV state for which the 10-minute averaged wind speed exceeds the onset wind speed of VIV or is close to the lock-in range.

2.2. A small VIV dataset

Similar to the previous subsection, a workflow to construct ML models for S2 is illustrated in Fig. 2. Note that S2, in which a few VIV data-points exist, is the most common case in real practice. In this scenario, the classification method (supervised learning) and clustering analysis (unsupervised learning) are introduced to predict VIV. As mentioned earlier, the methods introduced in Fig. 1 are also applicable when a VIV dataset is available.

A classification model can be trained using two classes of non-VIV

and VIV; however, an imbalanced dataset may inhibit the proper prediction of VIV, particularly when it shows poor performance for the minority class of VIV. This is because even though the classification model predicts only the majority class regardless of the input dataset, it nonetheless performs well in terms of the prediction accuracy. Similar to the classification model in S1, a Gaussian-mixture model is introduced to address imbalance problems via a synthetic oversampling of the minority dataset of VIVs.

The aim of clustering analysis is to partition a dataset into distinct clusters in terms of similarities. By doing so, dissimilarities between the data-points in the same cluster tend to be smaller than those in the other clusters. As described in the previous subsection (Stage 1 in Fig. 1), manual labeling demands a huge workload and entails inevitable error. When limited amounts of labeled data are available or it is difficult to guarantee the preprocessed labeling data, clustering analysis is introduced to address the issue. A clustering algorithm first groups datasets into several clusters. Since data-points in the same cluster share similar characteristics, each cluster is determined as either a VIV or a non-VIV class based on a small amount of guaranteed labeled data. For

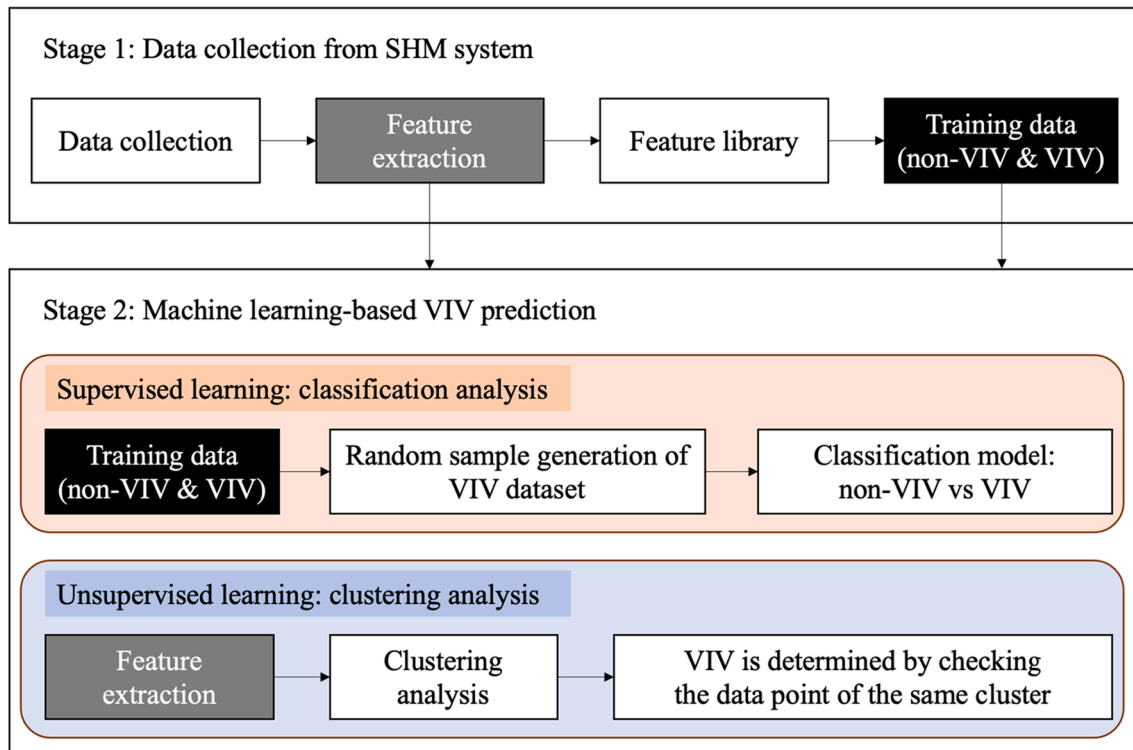


Fig. 2. ML-based VIV prediction when a small VIV dataset is available.

instance, if a cluster contains a fair amount of guaranteed labeled VIV data, such a cluster is determined as a VIV class. Once the clusters are annotated as non-VIV or VIV classes, it is possible to determine whether the newly collected dataset is VIV or non-VIV without manual labeling. If the same amount of guaranteed VIV and non-VIV data-points belong to the same cluster, either additional constraints or different clustering algorithms are introduced to address the issue.

3. Investigated bridge and dataset

3.1. Investigated bridge

The Jindo Bridge, a twin-cable stayed bridge in South Korea, is introduced to examine the proposed ML-based framework in Section 2.

The first section of the bridge was preliminarily constructed in 1984, while the second section was opened in 2005 to cope with the increased traffic volume. The two bridges have similar layouts, dimensions, cross-sectional shapes, and dynamic properties, as depicted in Fig. 3. VIVs have been occasionally reported in the second in-service Jindo Bridge, hereafter referred to as Bridge 2, and a series of wind tunnel tests showed that the interference effect between parallel decks magnified the VIV amplitudes [14]. Indeed, the past research efforts [12,25,26] also reported that twin-box girders and parallel twin bridges both are more susceptible to VIVs compared with a typical long-span bridge due to the gap vortex between upstream and downstream decks. After observing a VIV that exceeded the serviceability limitations in 2011, a multiple-tuned mass damper (MTMD) was installed at the center of the mid-span. Kim et al. [11] reported that the damping ratio for the first



Fig. 3. Parallel twin Jindo Bridge.

vertical mode was increased from 0.29 to 3.30%; accordingly, no significant VIVs have been reported since the MTMD was activated. Note that the previous studies on the VIV of Bridge 2 found the onset velocity to be about 10 m/s and the lock-in range was about 10–14 m/s [11,27].

3.2. Monitoring system

The investigated bridge has built-in sensors that include two accelerometers, an ultrasonic anemometer, and thermometers at the center of the main span, as shown in Fig. 4. It should be noted that those data were acquired after the installation of the MTMD. Also, weigh-in-motion (WIM) systems have been in operation at the bridge entrance to collect the following information: vehicle types, gross vehicle weights (GVW), vehicle lengths, vehicle classes, and passing times. The number of vehicles, the ratio of heavy trucks, and the total GVW during 10-minutes were extracted from the WIM data. In total, six different features (RMS acceleration, wind speed, temperature, number of vehicles, ratio of heavy trucks, and total GVW), measured for three months from June 1, 2019, to August 5, 2019, were used in this study. Since the dataset is preprocessed every 10 min, the total number of points was 13,968 ($= 97 \times 24 \times 6 = \text{days} \times \text{hours} \times 10 \text{ minutes}$).

3.3. Datasets

Fig. 5 shows the 10-minute average for velocity-RMS acceleration (V-A) curves during the monitoring periods. The color of each marker represents the number of passing vehicles. A quadratic formula between the RMS acceleration and the wind speed was found, which matched well with our knowledge of wind engineering. However, there are three notable regions in Fig. 5 that are worth investigating. To highlight the regions, they are respectively shaded in gray, red, and blue colors.

First, the data-points in the gray-shaded region show high levels of vibrational amplitude even though the mean wind speeds were not substantial. Since there was a high positive correlation between the number of vehicles and the RMS acceleration, these data-points were considered to be traffic-induced vibrations. Second, there are two groups of data-points where the mean wind speeds exceeded a certain threshold value, which is usually denoted as the critical wind speed of VIVs. The mean wind speeds of the data-points under the two regions were significantly large and within the lock-in range of the investigated bridge [14]. However, the RMS accelerations of the data-points in the red-shaded area follow a quadratic trend while those in the blue-shaded area are located outside this trend. To thoroughly understand the two different data groups, the time series of 10-minute RMS accelerations, the mean wind speed, the number of vehicles, and the temperature were examined, as shown in Fig. 6. As shown, large vibrations (depicted in the

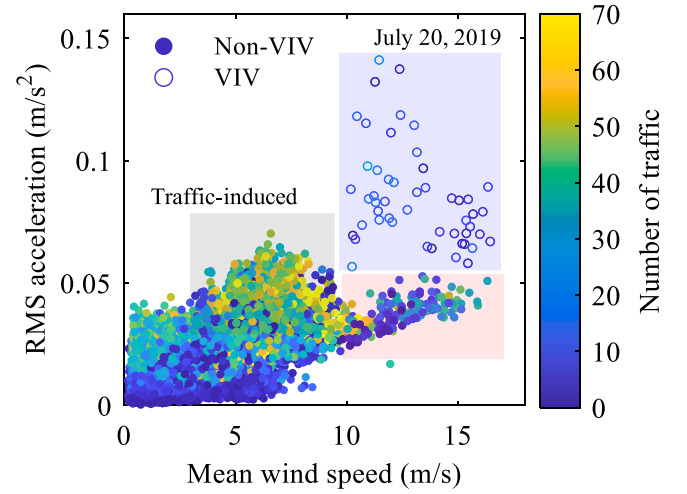


Fig. 5. Velocity-amplitude curve of the Jindo Bridge from 2019-06-01 to 2019-09-05.

red-shaded area) suddenly appeared on July 20, 2019. Such data-points correspond to the points under the blue-shaded area in Fig. 5. This is because when a typhoon is expected to pass the bridge the MTMD is typically closed in order to prevent an over-stroke of the moving masses stemming from the large buffeting responses. Based on this investigation and the fact that Typhoon Danas went through the Korean peninsula at the end of July 2019, we annotated the data-points in the blue-shaded area as a VIV dataset and others including data-points in the red-shaded area were considered a non-VIV dataset. Fig. 7 shows the representative examples of time histories and power spectral density functions of vertical acceleration at the center of the mid-span for the VIV and non-VIV datasets.

As previously mentioned, 50 data-points were labeled as VIVs, while the remaining 13,918 data-points were treated as non-VIV data-points. Again, this confirms that the number of VIV data-points is significantly smaller than that in non-VIV datasets. To properly demonstrate and delineate the proposed framework using the monitoring datasets, we divided the datasets into two groups: D1 and D2. Specifically, D1 refers to one-class data, and consists only of the normal state (i.e., non-VIV), while D2 refers to extremely imbalanced data that includes both VIV and non-VIV. In this regard, the monitoring data measured from July 17 to July 23 are referred to as D2 and the remainders as D1. Note that the number of D2 data-points was 1,007 with 50 VIV data-points, whereas that of D1 was 12,961 with no VIV data-points.

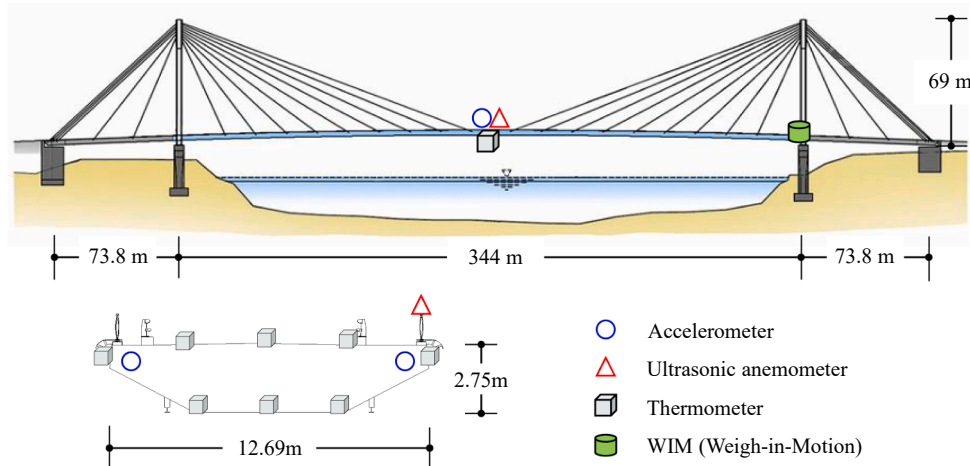


Fig. 4. Configuration of built-in monitoring sensors used in this study.

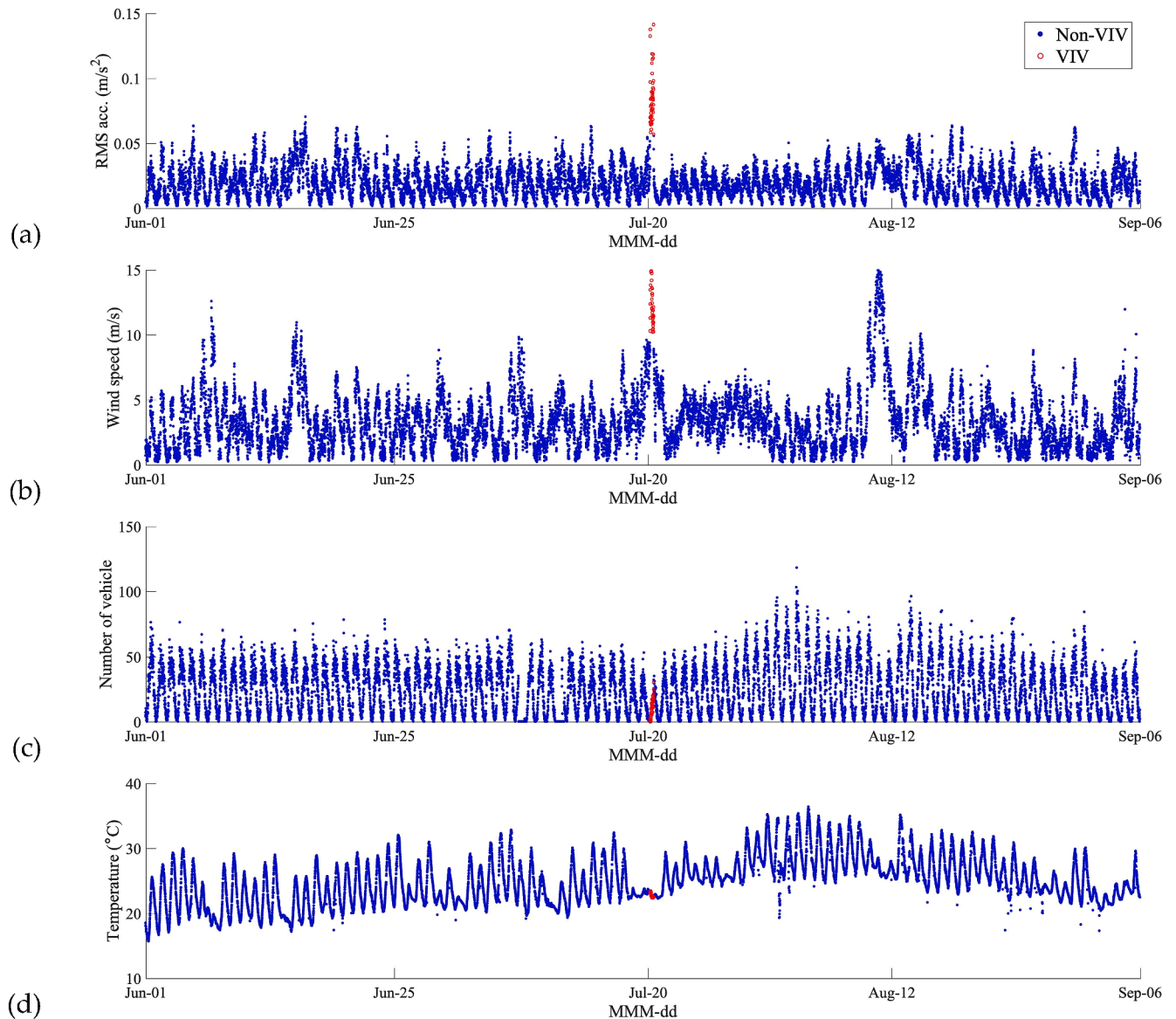


Fig. 6. Time histories of monitoring data: (a) RMS acceleration (m/s^2), (b) wind speed (m/s), (c) traffic volume, and (d) temperature ($^{\circ}\text{C}$).

4. Numerical investigations

The proposed ML-based VIV prediction frameworks were demonstrated using the measured datasets described in Section 3.3. As shown in Section 2, three strategies were introduced for S1 and two for S2. Different ML algorithms were implemented for each strategy to verify the general applicability of the framework.

4.1. S1: No VIV dataset

Based on the framework proposed in Section 2.1, each ML model was trained using only D1. The performance of trained ML models was verified based on D2. Note that D2 was only used for verification purposes and was not employed in training. Although post-processing (Stage 3 in Fig. 1) is required to determine the VIV state following predictions using the ML models as described in Section 2.1, the numerical investigations for S1 is focused mostly on developing the ML models, i.e., Stage 2 in Fig. 1. As mentioned earlier, the VIV state can be determined by assessing the EOCs of the predicted non-normal state. However, further study is needed to construct an automated and efficient framework to identify the VIV state from the non-normal state

dataset.

4.1.1. Supervised learning: Regression analysis

Various regression analysis algorithms have been developed to predict continuous output variables for given input values. Because a Gaussian Process Regression (GPR) model not only predicts the mean trend of the output but also provides its confidence interval (CI) [28], GPR was selected to predict VIV for given EOCs. In addition, considering the fact that the dynamic responses of the bridge deck should be predicted in terms of probabilistic approaches due to the high degree of uncertainties in the structural, environmental, operational, and measurement conditions, the GPR is a proper candidate for this investigation.

The training of the GPR model starts with selecting a proper kernel function that can describe the covariance matrix of the dataset. Based on the Bayesian optimization algorithm, an Automatic Relevance Determination (ARD) rational quadratic kernel was selected from among widely used kernel functions such as squared exponential, exponential, Matern, rational quadratic, and ARD. The GPR model was trained to predict the RMS acceleration of the bridge deck for the given five features of EOCs. To check the over-fitting issue, 80% of D1 was selected as

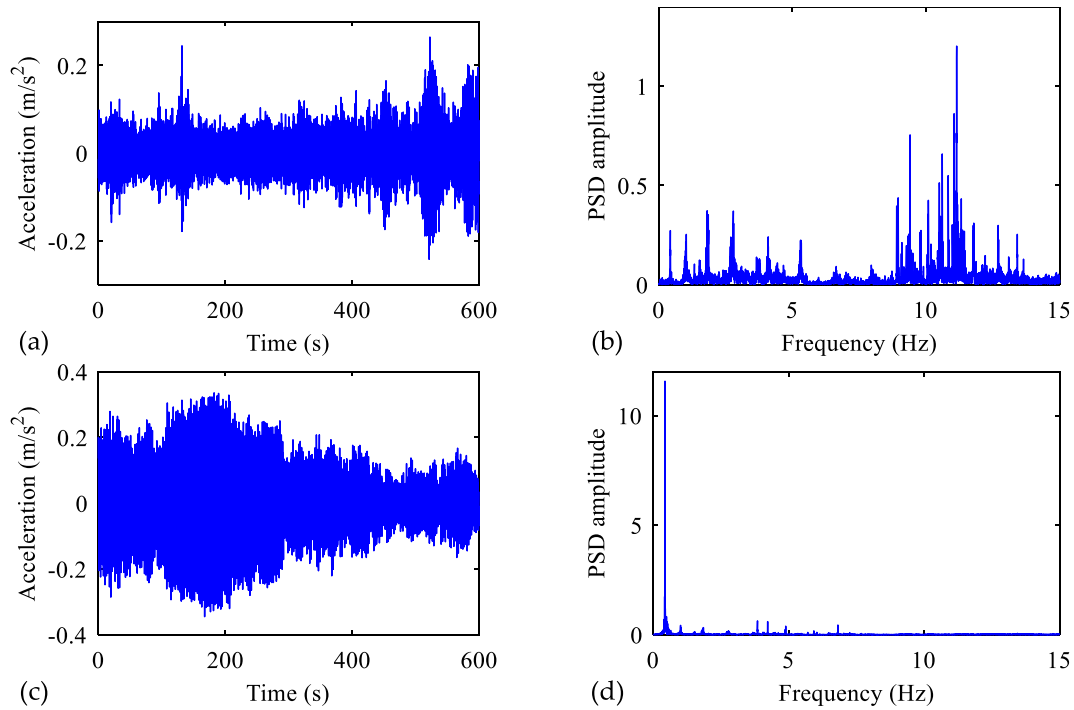


Fig. 7. The representative examples of vertical acceleration: (a) time history and (b) PSD function of VIV ($U = 12.11$ m/s), and (c) time history and (d) PSD function of non-VIV ($U = 5.33$ m/s).

a training dataset, while the remaining 20% was used for testing following the training.

Before performing the training, it was important to investigate the characteristics of the dataset in order to achieve a considerable prediction performance. Fig. 8(a) illustrates a histogram of the training dataset (i.e., 80% D1) along with RMS acceleration values, which shows a large skewness. It is widely known that a skewed or distorted dataset could hinder proper pattern recognition during training. To address the skewness problems in the training dataset, a synthetic minority over-sampling technique (SMOTE) was employed [29]. The SMOTE uses the k -nearest neighbor approach to duplicate the samples of the minority classes, which can amplify each bin of the histogram. Fig. 8(b) uses the SMOTE to demonstrate a histogram of the augmented dataset.

Using the augmented dataset and the selected kernel function, the 80/20 holdout procedure was repeated 100 times to validate the stability of the model against data partitions. In the 80/20 holdout procedure, 80% of the dataset was randomly selected to train the ML model, while the remaining 20% was used to validate the trained model. Fig. 9 shows an example of the measured and predicted time series of RMS acceleration with the test dataset. This shows that the predicted values

properly followed the mean trend of the test dataset. In particular, 96.75% of measured values were located within 99% of the CI (yellow area). Moreover, the averaged statistical indices of 100 repeated validations were achieved as a root mean squared error (RMSE) of 0.00687 m/s² (coefficient of variation (c.o.v) = 2.04%) and a correlation coefficient (R) of 84.91% (c.o.v = 0.76%), which confirms the robust performance of the GPR model in forecasting dynamic responses under normal operation.

The GPR model combined with a relaxed rule was employed to identify VIVs. The relaxed rule determines the VIVs of the i -th data-point as the summation of the number (n) of consecutive events, including the i -th data-point, that exceed the summation of the corresponding upper limits of the CI. A mathematical expression of the relaxed rule is written as Eq. (1).

$$VI_i = \begin{cases} 1, & \text{for } \sum_{i-n/2}^{i+n/2} y_i > \sum_{i-n/2}^{i+n/2} CI_i \\ 0, & \text{for } \sum_{i-n/2}^{i+n/2} y_i < \sum_{i-n/2}^{i+n/2} CI_i \end{cases} \quad (i = 2, 3, \dots, N-1) \quad (1)$$

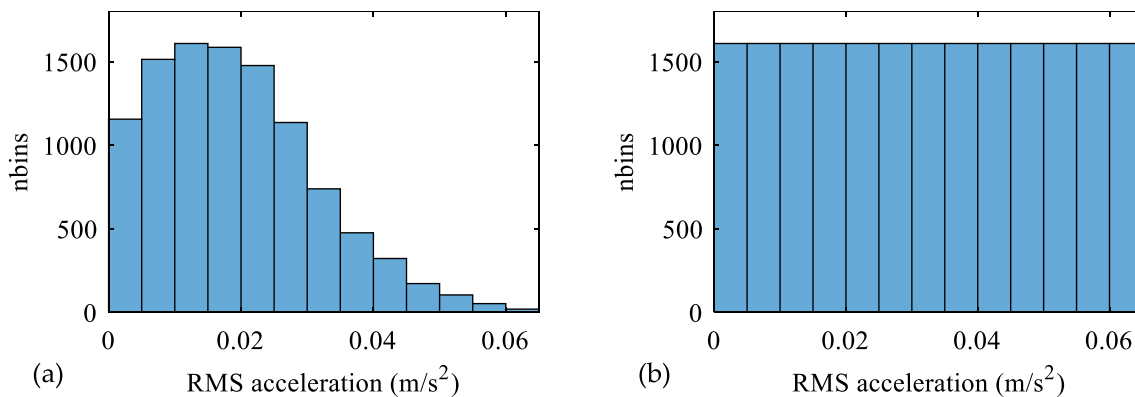


Fig. 8. Histogram of the training dataset (a) original dataset, and (b) augmented dataset using the SMOTE.

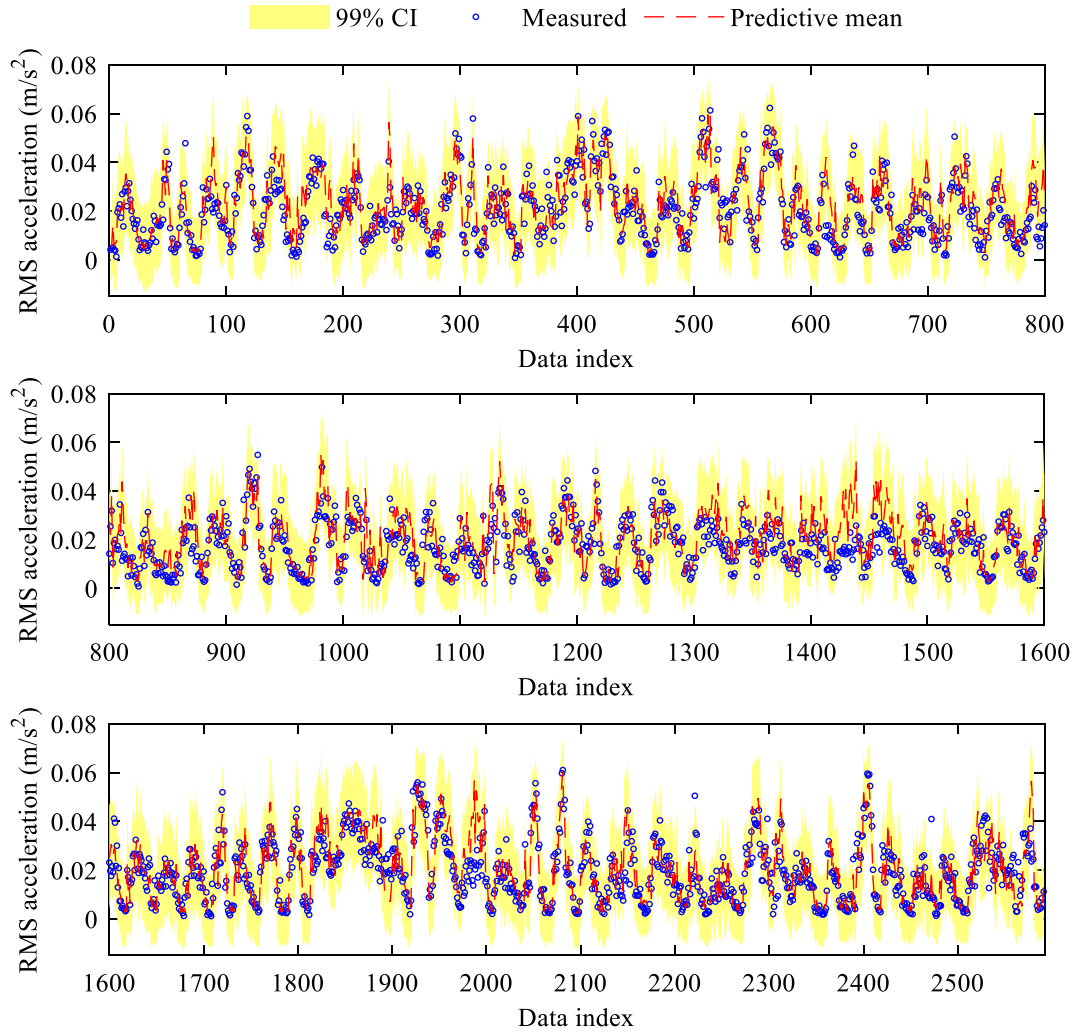


Fig. 9. Predicted mean trend and its confidence interval from the GPR model for the test dataset.

In Eq. (1), VI_i stands for the VIV identification index for the i -th data-point; y_i denotes the i -th predicted value; CI_i is the i -th upper boundary of the CI, and N is the total number of data-points. Comparative studies were carried out to determine the hyperparameters of the relaxed rule. To this end, five different numbers of consecutive events, 1, 3, 5, 7, and 9, and two CIs, 95% and 99%, were employed. The optimal hyperparameters were determined by comparing the micro-F1 [30] of D1 and

D2 with a different n and CI, as shown in Fig. 10. Overall, (1) the micro-F1 exceeded at least 98.5% in all cases; (2) a 99% CI outperformed a 95% CI; (3) slightly more misidentifications appeared in D1 than in D2, but these were negligible; (4) the micro-F1 of D2 was 1.0 for an n of 3 and 5; and (5) the micro-F1 of D1 gradually increased with a larger n , indicating a more relaxed inspection. Meanwhile, the obtained results demonstrated the robustness of the parameter selections for VIV

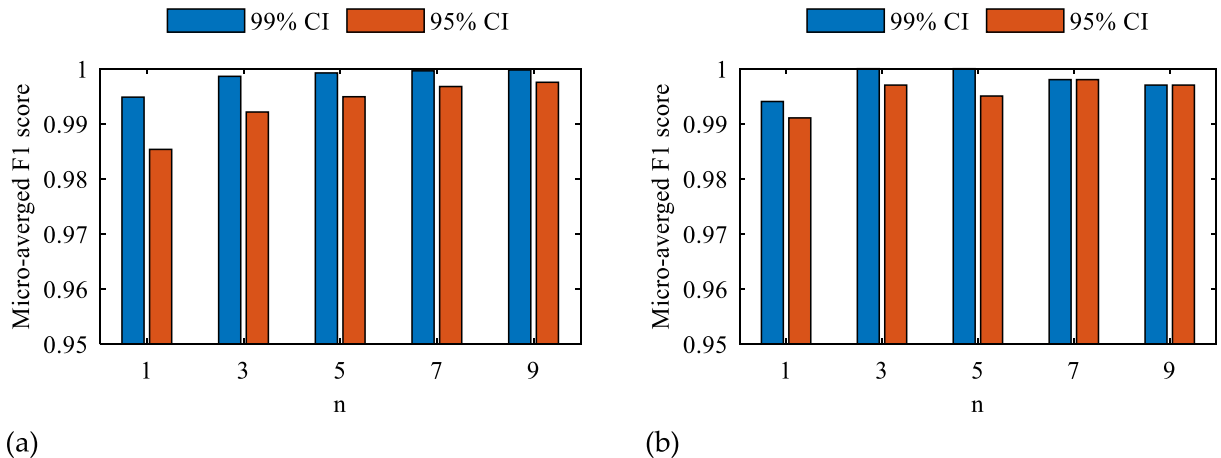


Fig. 10. Micro-F1 with the different hyperparameters: (a) D1 and (b) D2.

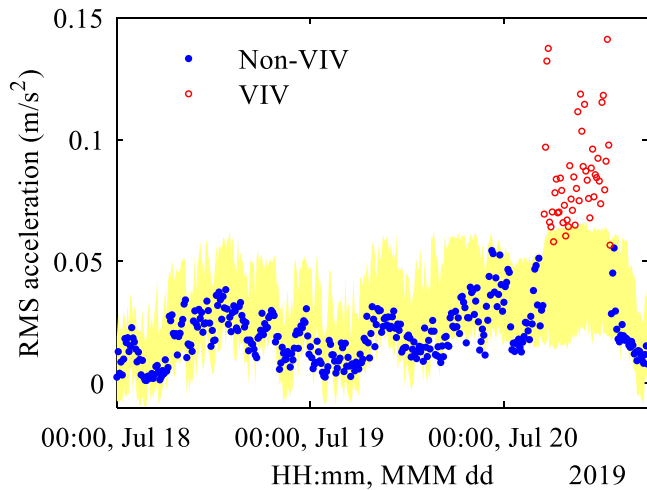


Fig. 11. Identification of VIV from the GPR model for D2 with $n = 5$ and CI of 99%.

identification, and the n of 5 and CI of 99% proved to be the best hyperparameters for the relaxed rule.

Fig. 11 shows the results of VIV identification using an n of 5 and a 99% CI for D2. There was good agreement between the identified VIV from the GPR model and the labeling of data using engineering judgment. The time history shows that the anomaly was only detected temporarily and did not last any longer, indicating that these anomalies were not caused by other permanent non-normal states such as damages or structural degradation. In addition, the 10-minute averaged wind speed of the anomalies ranged from 10.17 to 16.43 m/s, which exceeded the onset wind speed and approximated the lock-in range of Bridge 2. These findings confirm that such detected anomalies are VIV events. The confusion matrix of GPR-based VIV identifications for D1 and D2 are tabulated in Table 1 ($n = 5$ and 99% CI). Even though a few false identifications were observed for D1, the results demonstrate the effectiveness and robustness of the proposed framework in identifying VIVs.

4.1.2. Supervised learning: Classification analysis

According to the framework in Section 2.1, applying classification ML algorithms to a dataset with only one class (i.e., non-VIV in our problem) requires the same number of abnormal-state datasets. To this end, a Gaussian mixture-based probability-density function (PDF) was constructed for a non-VIV dataset. Here, the term non-VIV dataset refers to a normal-state dataset. In developing the PDF, the optimal number of Gaussian mixtures was determined by evaluating the performance of the constructed PDF. With the normal-state dataset, 80% was used to fit the PDF, while the remaining 20% was used for assessment. The mean log likelihood value of the PDF for the remaining 20% of the normal-state dataset appears in Fig. 12 along with the number of mixtures. As shown in the figure, no significant performance improvement was observed for a number of components greater than or equal to 4. Thus, 4 mixtures were used for the data augmentation.

The data-points of the abnormal state were randomly generated from the PDF. The abnormal state was determined by comparing its probability density to a certain value. In this study, we determined the

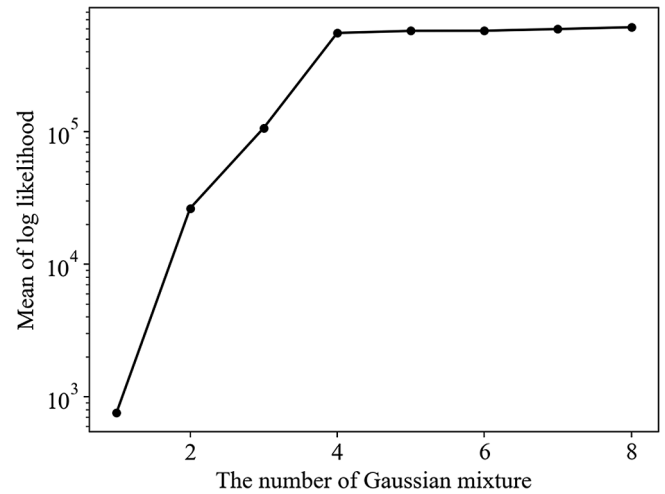


Fig. 12. Mean of the log likelihood along with the number of components for the 20% normal state dataset.

threshold value to be 0.5 percentile of the log likelihood of the normal-state dataset. By following the procedure, 12,961 abnormal-state data-points were generated.

By using the normal-state and abnormal-state datasets (a total of 25,922 dataset), it was possible to construct a classification model. Among various classification algorithms, a deep neural network (DNN) model was used. The architecture of the DNN model is shown in Fig. 13. The model has 4 hidden layers and adopts a rectifier linear unit (ReLU) as an activation function [31]. To classify the normal and abnormal states, a sigmoid function was employed in the last hidden layer and binary cross entropy was selected as a loss function. The number of each unit in the layers was estimated using Bayesian optimization. To check the over-fitting problem, we randomly selected 80% of the total dataset to train the DNN model, while the remaining 20% was used to validate the performance of the DNN model. The DNN model was trained for 10,000 epochs with a batch size of 512 using an Adam optimizer [32]. The levels of prediction accuracy for the train (20,738) and validation (5,184) datasets were both 99%. Thus, we confirmed that the classification model was not over-fitted to the training dataset and successfully extracted the general pattern between input and output.

Both the D1 and D2 datasets were employed to demonstrate the performance of the trained DNN model. Table 2 charts the confusion matrix of the DNN model for D1 and D2. Note that in this problem, an abnormal state was considered VIVs. Since 0.5 percentile of the likelihood of a trained dataset was assumed to be the threshold value in the data augmentation technique, some of the non-VIV data-points were classified as abnormal-state datasets. Even though no VIV dataset was used in training, however, the DNN model showed promise by classifying the VIV dataset as an abnormal state.

To further elaborate on the performance of this framework along with the threshold value in the data augmentation procedure, an augmented dataset was regenerated from the PDF by setting the threshold value as 0.1 percentile of the likelihood instead of the 0.5 percentile used in the previous investigation. In other words, only extremely rare data-points were assumed to represent an abnormal state. A DNN model was trained using the same architecture and training

Table 1

Confusion matrix of the GPR for D1 and D2 ($n = 5$, 99% CI).

	D1		Predict			D2		Predict	
	Normal	Non-normal	Normal	Non-normal		Normal	Non-normal	Normal	Non-normal
Actual	Normal	Non-normal	12,951	10	Actual	Normal	957	0	
			0	0		Non-normal	0	50	

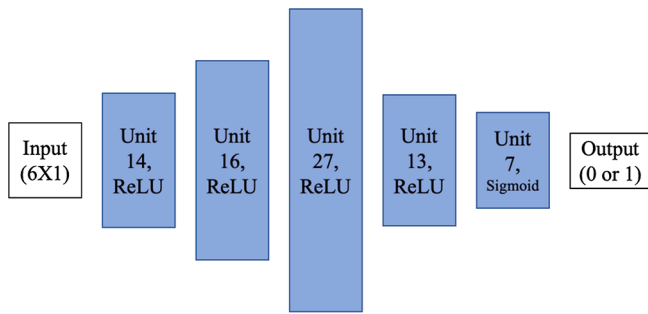


Fig. 13. Architecture of the deep learning-based classification model.

environment. The confusion matrix using the newly trained model appears in Table 3. We found that a threshold value of less than a certain limit would not improve the prediction performance of the classification model. Since the augmented dataset was based only on the non-VIV dataset, it was difficult to distinguish the data-points located at the edge of the fitted Gaussian mixture-based PDF.

The aforementioned limitation was investigated using the scatter plot shown in Fig. 14, which is a scatter plot of D2 in terms of the wind speed and RMS acceleration. The gray and red hollow circles represent the normal and abnormal data-points, respectively. Moreover, the blue and black 'x' marks represent the false positive and false negative cases, respectively, where the false positives incorrectly indicate the normal state, while the false negatives are the opposite. Note that the prediction errors presented in Table 2 are used. As shown in the figure, the incorrectly predicted data-points are located at the boundary of the normal dataset. In particular, the false positive cases are the two closest points from the center of the abnormal dataset among the normal data-points. This investigation showed the inevitable existence of prediction errors, particularly near the classification boundary when using this framework.

4.1.3. Unsupervised learning: Autoencoder

To predict VIV using an autoencoder, VIV data-points were classified as anomalies. In other words, an autoencoder model was trained using a dataset with only non-VIV data-points, and the model performance was then predicted using a test dataset with both VIV and non-VIV data-points. To this end, the architecture shown in Fig. 15 was used. A pre-processing technique was employed to normalize the data that lay between 0 and 1. To check the impact of the latent variables on the prediction accuracy, 5 different autoencoder models were trained by varying the latent variables from 1 to 5. A mean squared error was adopted as the loss function, and an Adam optimizer was employed to train the autoencoder models.

After 10,000 epochs of training batch sizes of 512, the performances of the five trained models were evaluated using the D2 dataset. Different threshold values were assigned to each autoencoder model to determine

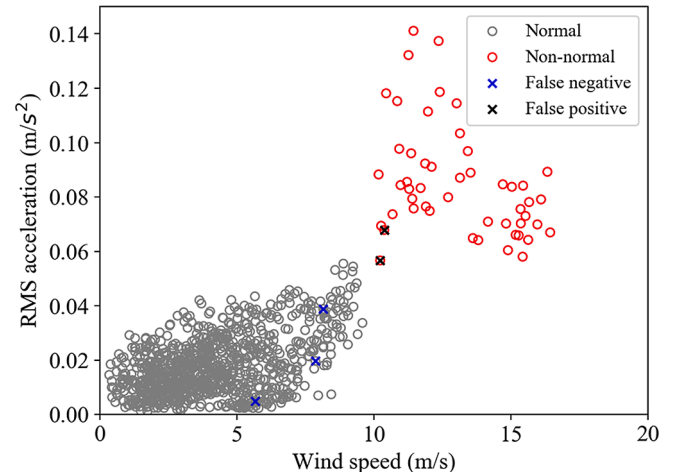


Fig. 14. Scatter plot of the D2 dataset (hollow circles) and the incorrect prediction points (x markers).

the states of anomaly. The values were assigned in terms of the reconstruction error of the training dataset to insure that use of the autoencoder model would result in the smallest rate of prediction error for the D2 dataset. The threshold values are presented in Table 4 along with the number of false predictions. Note that false positives involve cases where a prediction result incorrectly indicates a non-VIV data-point, while the false negatives are the opposite. As shown in the table, the performance of the autoencoder model with two latent variables was superior to that of the other autoencoder models.

Using the autoencoder model with two latent variables, the reconstructed vectors of D2 were plotted in terms of the wind velocity and RMS acceleration in Fig. 16(a) using the original dataset in Fig. 16(b). As the figure shows, since only two latent variables were used, the decoder could not perfectly reconstruct the dataset. The original dataset in Fig. 16(b) shows that the wind speed and RMS acceleration of the non-VIV data-points are located within a range of less than 10 m/s and 0.06 m/s², respectively. Since the autoencoder model was trained with only the non-VIV dataset, the decoder could not properly reconstruct the VIV data-points, particularly those for RMS acceleration.

Based on this investigation, in addition to the reconstruction error, a

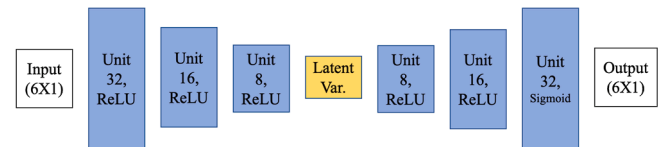


Fig. 15. Architecture of the autoencoder model.

Table 2

Confusion matrix of the classification model for the D1 and D2 datasets.

D1		Predict		D2		Predict	
		Normal	Non-normal			Normal	Non-normal
Actual	Normal	12,946	15	Actual	Normal	953	4
	Non-normal	0	0		Non-normal	1	49

Table 3

Confusion matrix of the classification model for D1 and D2 when 0.1 percentile of the likelihood is selected as the threshold.

D1		Predict		D2		Predict	
		Normal	Non-normal			Normal	Non-normal
Actual	Normal	12,945	16	Actual	Normal	953	3
	Non-normal	0	0		Non-normal	2	48

Table 4

Threshold values and the number of false predictions for the 5 autoencoder models.

Latent variables	1	2	3	4	5
Threshold in terms of percentile	98	99.45	99.9	99.9	99.9
Number of false positive	7	3	6	1	1
Number of false negative	3	3	4	8	6

threshold for RMS acceleration was applied to facilitate a better prediction for the VIV data-points. This new threshold was defined as the proportion of the reconstruction error from RMS acceleration that is greater than $1/6$. Six different forms of information were used as input, and if the autoencoder models were properly trained, we expected the reconstruction error to be equally distributed to each input. Table 5 shows the results of the false prediction when applying the new rule. The performances of the autoencoder models were greatly improved compared with the previous results shown in Table 4, with the exception of the models with relatively greater numbers of latent variables. This is because a large number of latent variables led to the need to fully train both the global and local patterns of the input and output datasets, which creates difficulties for the autoencoder models to discern the VIV and non-VIV data-points located near the boundaries of two classes.

4.2. S2: Small VIV datasets

Both D1 and D2 datasets are available in S2. Based on the framework proposed in Section 2.2, two learning methods were introduced.

4.2.1. Supervised learning: Classification analysis

When the data-points of VIVs are available in a non-VIV dataset, it is possible to create a ML-based model that can classify whether a structure is under VIV or not. Non-VIV data-points, however, will generally make up a large portion of the measured data-points. To address this classification imbalance, a PDF was constructed to properly reproduce the VIV dataset. Of the VIV datasets in D2, 80% were used to fit a Gaussian mixture, while the remaining 20% were used to determine the optimal number of Gaussian mixtures. By comparing the mean likelihood value of the test dataset in the present study, 5 mixtures were selected as the optimal number that would fit the PDF. By using the PDF, the VIV dataset was used to augment the non-VIV dataset. During the data augmentation, data-points with negative values and RMS acceleration of less than 0.45 m/s^2 were discarded based on engineering judgment. The data samples from high wind speeds but relatively small RMS acceleration could be due to buffeting, but considering the objective of this study, such a dataset would not be employed in training. The scatter

plots of the non-VIV (D1) and the augmented VIV datasets are described in Fig. 17 in terms of the wind speed and RMS acceleration.

Using the combined datasets, a DNN model was trained to classify the non-VIV and VIV. Note that the D2 dataset was not used during training. By using the Bayesian optimization, the number of units for each layer was determined, as shown in Fig. 18. By setting the binary cross entropy as a loss function, an Adam optimization algorithm was employed to train the parameters. The training was carried out using 512 batch sizes for 10,000 epochs.

The confusion matrix of the trained DNN model for the D1 and D2 datasets is shown in Table 6. Since the VIV dataset was augmented from the Gaussian mixture-based PDF, it was possible to construct a DNN model that could perfectly classify the VIV and non-VIV data-points. The training revealed that the six features have sufficient information to discern VIV against a normal state. However, note that it is difficult to

Table 5

The number of false predictions for the 5 autoencoder models with an additional threshold.

Latent variables	1	2	3	4	5
Number of false positive	0	1	3	1	1
Number of false negative	3	3	4	8	7

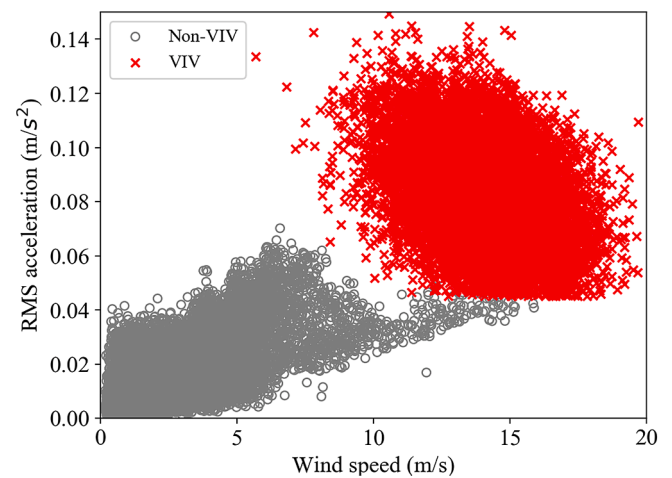


Fig. 17. Scatter plot of the D1 dataset (hollow gray circle) and the augmented VIV dataset (red x). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

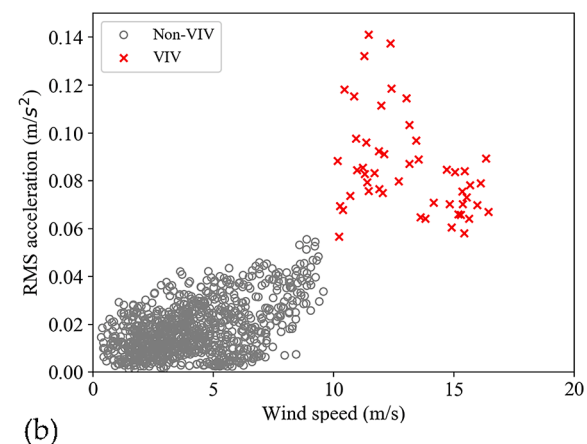
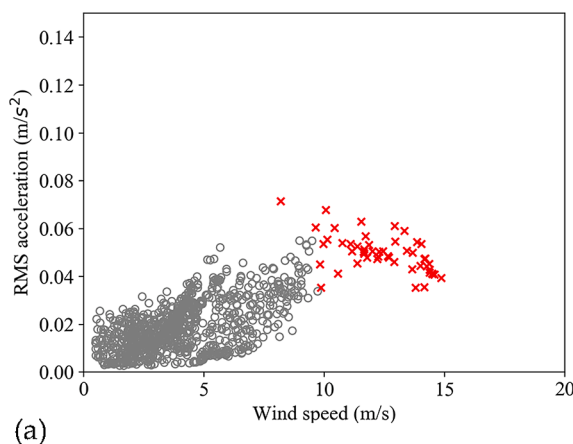


Fig. 16. Scatter plot of the D2 dataset in terms of wind speed versus RMS acceleration: (a) reconstructed vector from the decoder, and (b) measured data from sensors.

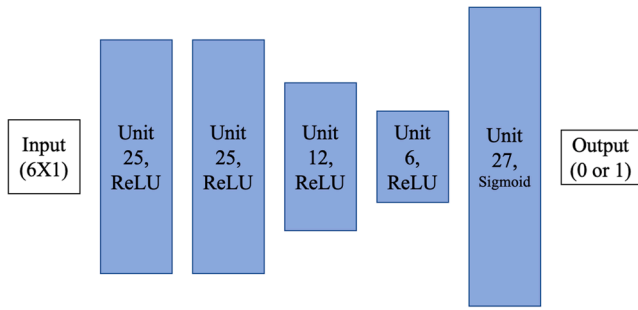


Fig. 18. Architecture of the DNN model.

Table 6

Confusion matrix of the DNN model used to classify both VIV and non-VIV for D1 and D2 datasets.

D1		Predict		D2		Predict	
		Non-VIV	VIV			Non-VIV	VIV
Actual	Non-VIV	12,961	0	Actual	Non-VIV	957	0
	VIV	0	0		VIV	0	50

find the patterns of the prediction for the data-points that were discarded in the data augmentation process because such data-points are not used in training. In such a case, an additional class for buffering response is required, thus further study is currently underway to address this issue.

To further examine the relationship between input and output, a sensitivity analysis was performed. To this end, only 5 of 6 input features were used to construct a DNN model that could classify both non-VIV and VIV. Since 6 features were introduced in this study, 6 different DNN models were developed for the sensitivity analysis. The DNN models were trained under the same training environment, and the prediction accuracy (the number of false predictions) of the six different DNN models is listed in Table 7. The results of this investigation showed that RMS acceleration, wind speed, and the number of vehicles are the most important features that could be used to predict VIV from among the six features, which matches well with our knowledge of wind engineering.

Based on the results from the sensitivity analysis, a new DNN model was trained using the following four features: RMS acceleration, wind speed, temperature, and the number of vehicles. Using the same training methodology and model composition, the new DNN model was developed and its prediction results appear in Table 8. As shown in the table, the DNN model continued to show good performance even though some information had not been used. This is because the number of vehicles, number of trucks, and total weight of all passing vehicles showed a high correlation, i.e., shared similar information.

Table 7

The number of false predictions using the six different DNN models.

Dataset	Excluded information					
	Number of vehicles	Number of trucks	Total weight	Wind speed	Temperature	RMS acceleration
D1	1	0	0	6	0	55
D2	0	0	0	0	0	0

Table 8

Confusion matrix of the DNN model with reduced features.

D1		Predict		D2		Predict	
		Non-VIV	VIV			Non-VIV	VIV
Actual	Non-VIV	12,959	2	Actual	Non-VIV	951	6
	VIV	0	0		VIV	0	50

4.2.2. Unsupervised learning: Clustering

When the labeling quality is unsatisfactory, clustering analysis is useful to predict VIV. It was noted, however, that a small amount of data-points requires proper labeling as either VIV or non-VIV in order to predict VIV. In this examination, it was assumed that 10% of VIV and non-VIV datasets were properly labeled. Among various clustering methods, *k*-means clustering [33] was employed for this problem. This method partitions a dataset into *k* clusters where each observation belongs to the cluster with the nearest mean. To find the optimal number of clusters, *k*-means clustering analysis is carried out by increasing the number of clusters. The sum of the squared distances of data-points to their closest cluster center was plotted along with the number of clusters, as shown in Fig. 19. The optimal number of clusters was defined as the value *k*, for which a decrement in the sum of the squared distance was less than 3% when *k* was increased to *k* + 1. Based on this criterion, an optimal *k* of 16 was selected.

By setting *k* = 16, *k*-means clustering was performed using the six features described in Section 3.3. The results of clustering analysis are shown in the format of a scatter plot in Fig. 20(a) for wind speed and RMS acceleration. Using the guaranteed labeled data, each cluster was determined as either VIV or non-VIV. For classification, a cluster that contained no labeled data was merged with the cluster that was most similar and had already been annotated. By following this procedure,

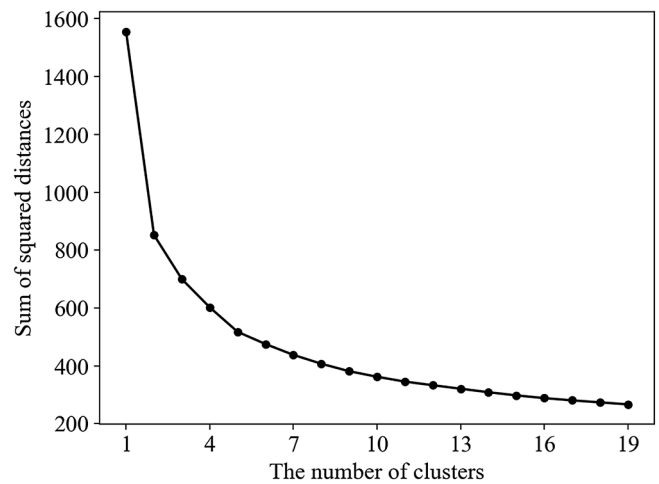


Fig. 19. Sum of squared distances along with the number of clusters.

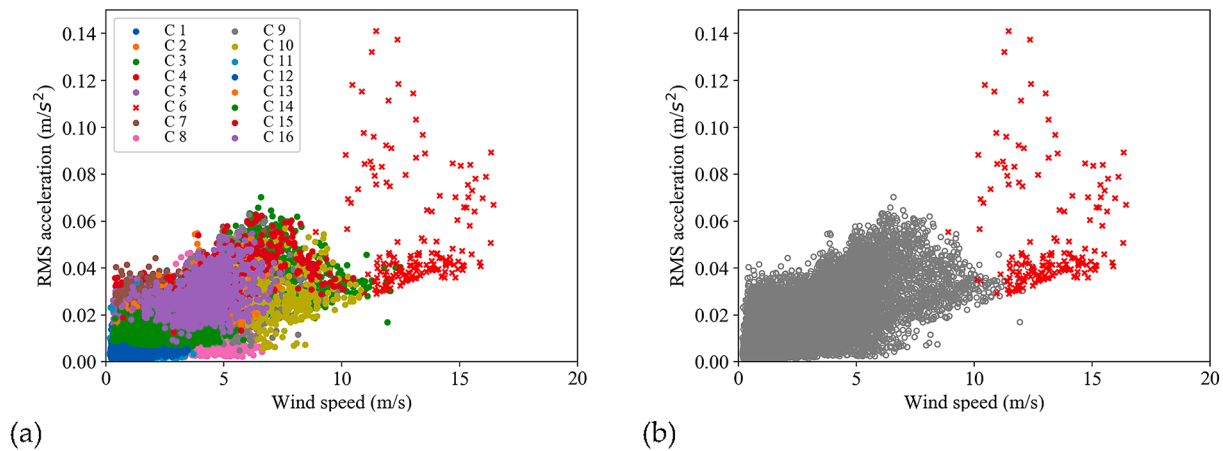


Fig. 20. Scatter plot of the D1 and D2 datasets along with results of clustering analysis.

Table 9

Confusion matrix for the D1 and D2 datasets based on clustering analysis.

	D1		Predict			D2		Predict	
	Non-VIV	VIV	12,842	119		Non-VIV	VIV	956	1
Actual	Non-VIV	VIV	0	0	Actual	Non-VIV	VIV	0	50

only Cluster 6 (red $\times$ marker in Fig. 20(a)) was defined as a VIV cluster and is highlighted in Fig. 20(b) by highlighting other clusters with gray hollow circles. Using the analysis results, which are similar to that of a k-nearest neighbors (KNN) algorithm, it was possible to identify the clusters as either VIV or non-VIV by comparing the distance between the points and the center of each cluster.

Using the identified clusters, the confusion matrix could be constructed, as shown in Table 9. The table shows that the clustering analysis-based VIV prediction demonstrated good prediction accuracy for D2, but some false negative results appeared in D1. This is because clustering analysis placed the data-points showing a wind speed greater than about 10 m/s into the same group (both red and blue shaded areas in Fig. 5). If additional constraints are imposed with the clustering analysis, better prediction results could be expected.

4.3. Comparison of each learning method

In the previous subsections, a total of 5 different ML models were introduced to predict VIV with promising results in terms of prediction accuracy. The proposed framework was helpful particularly when the number in the VIV dataset was relatively low. This subsection summarizes the procedure and introduces insights that researchers should be aware of when introducing each learning algorithm. First, the regression analysis in S1 can be used to numerically predict the RMS acceleration. VIVs are determined by comparing the error between the prediction and the measurement under an assumed CI. The numerical investigation showed good prediction accuracy, but to achieve better prediction performance, care must be taken in proposing a determination rule, as when we proposed a relaxation rule in Section 4.1.1. Second, the classification model in S1 was successfully used to classify the abnormal state apart from the normal state (non-VIV). The abnormal-state dataset was used to augment the normal-state dataset via the use of a PDF designed to fit the normal-state. Even though no VIV dataset was used in training the classification model, a high level of prediction accuracy was gained. However, the classification model could not efficiently classify the two states for data-points located near the boundaries of the two classes. Moreover, due to the nature of the data augmentation, the abnormal state does not always guarantee the presence of VIV, which

could lead to false predictions in future applications. Third, an autoencoder was used to predict VIV by assuming VIV existed as an anomaly in the dataset. Reconstruction error from a decoder was used to predict VIV. Similar to regression analysis, fine-tuning is required to obtain good prediction accuracy. Fourth, in order to use the classification analysis in S2, a VIV dataset was first used for augmentation in order to address the issue of an imbalanced dataset. Since the method used the majority of information among the five strategies, the classification model produced the highest prediction accuracy in this study. Finally, when labeling is not properly carried out, clustering analysis is a good strategy to predict VIV. As described earlier, however, a few data-points should be properly labeled before employing clustering analysis. Even though the classification relies on a small number of labeled data-points, the method attained good prediction performance in discriminating between the

Table 10

Summary of ML-based VIV prediction.

Scenario	Learning methods	Descriptions	Remarks
S1	Probabilistic regression analysis	RMS acceleration is predicted. VIV is determined by comparing the error between the prediction and the measurement with CI.	Proper determination rule is required
	Classification analysis	Data augmentation procedure is introduced for the abnormal-state dataset. Normal state (non-VIV) and abnormal state are classified.	Inevitable prediction error is observed in classification boundary
	Autoencoder	VIV is determined by comparing the reconstruction error with an assumed threshold value.	Care is needed to determine the threshold
S2	Classification analysis	Data augmentation procedure is introduced for the VIV dataset. Non-VIV and VIV dataset are classified.	Superior performance is observed because the majority of given information is employed
	Clustering	Each cluster is labeled as non-VIV or VIV class based on a few numbers of guaranteed dataset. Non-VIV and VIV are determined according to KNN algorithm.	The better performance is expected when the method is combined with other ML models

VIV and non-VIV states. If clustering analysis is combined with other machine-learning methods, better prediction accuracy is expected. A description and remarks concerning the framework are summarized in Table 10.

5. Conclusions

A general framework for introducing machine-learning (ML) algorithms to predict vortex-induced vibrations (VIVs) is proposed in this study. By assuming two practical scenarios along with the existence of VIV datasets, a total of five different strategies are proposed using various ML algorithms. To attest to the applicability and effectiveness of the proposed framework, actual monitoring data from a cable-stayed bridge in South Korea was employed. Six important features were extracted from the monitoring data, which was used in developing ML models. However, note that the applicability of the framework did not depend on the data features. Based on the numerical application, the proposed framework efficiently addresses the use of insufficient information in predicting VIV. Moreover, to attain insight into each ML strategy in predicting VIV, further numerical examination was carried out. Although all strategies showed good performance in discriminating between VIV and non-VIV, we found that when more information is employed in developing the ML model, better prediction results are obtained. The findings from this study can be used in the near-real-time operation and maintenance of the serviceability of bridges, which could enhance the disaster resilience of long-span bridges that are subjected to wind forces.

Despite successful implementation, the following three limitations exist. First, in order to use the proposed framework, data preprocessing (e.g., labeling) is required based on engineering experience and judgment. Second, a systematic procedure is required to identify the VIV state from the predicted non-normal state dataset with a minimum of user intervention when no VIV dataset is employed in training the ML models. Third, different ML models are needed for each bridge. In other words, it is difficult to guarantee the prediction performance of the ML models developed for the Jindo Bridge when they are used to predict VIV for other bridge systems. In this regard, further studies are currently underway to accomplish the following three points: (1) to propose an automated data preprocessing technique that includes labeling, annotating, data cleaning, and pre-classification; (2) to characterize various wind-induced vibrations in terms of environmental and operational conditions without relying on ML models; and (3) to develop a universal VIV prediction ML model that could be applied to various bridge systems without further training.

CRedit authorship contribution statement

Sunjoong Kim: Conceptualization, Methodology, Investigation, Validation, Formal analysis, Data curation, Software, Visualization, Funding acquisition, Writing – original draft. **Taeyong Kim:** Supervision, Methodology, Investigation, Validation, Formal analysis, Software, Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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